DIGITAL POWER

Seniority Guidelines: Data Engineering

V1.0

Trainee

- Experience with python programming: Proficient use of functions, knowledge of data structures
- Interested both fields of data and IT
- Should have the capacity and talent to develop themselves to a junior data engineer in 3 months

Junior

- We take the approach of a junior ETL developer
- Knowledge about data types (Strings, Floats, Integers, Booleans)
- Able to work with and understand Data Models
- Should be able to write code for transformations and access (read/write) to any storage systems
- Able to create virtual python environments
- Consultant is familiar with basic concepts like OOP, DTAP and DRY, should be able to write functioning, clean code by themselves
- Knows how to work with git (dev branching, merge, solve merge conflicts) + existing CICD pipelines
- Be able to find and solve basic bugs using an IDE debugger
- Abstract knowledge about cloud technology

Medior

- Knows their way around python well:
    - Object oriented programming
    - Be able to write efficient code:
        - Maintainability
        - Complexity
        - Performance
        - Reusability
        - Error handling
    - Tests
    - Logging
    - Setup coding standards PEP8, Linters, Formatters
- CICD:
    - Understand all of the steps of a CICD
    - Know how to update / setup a basic CICD
- Should be aware of all the data components in a Data Project/Platform
- Is experienced in one or more cloud environments.
- Be able to setup at least one storage component on the cloud, on simple setups:
    - Object stores
    - Relational DBs
- Has experience with tools like bash scripting, docker and/or SQL
- Has experience with cloud technologies like data warehouses, data lakes, orchestrators, distributed processing and/or analytical processing
- It is okay, to compensate some of these hard skills with other related to the domain

Senior

- Has developed a lot of in-depth knowledge or has a strong all-round profile.
- Should have the capacity to take on the role of lead developer.
- Not every senior DE needs to have outstanding soft skills. However, we do expect them to have clear communication skills towards business and other developers.
- Be able to design and implement most of the components of a small-medium size data platform considering the following aspects:
    - Orchestration
    - Compute engines: Distributed / Centralized (with concurrency)

- Storing
- CICD
- Security (basic concepts and share responsibility models according to the cloud providers)
- User management
- Infrastructure as a code
- Data modelling